Fares Fawzi

Lausanne, Switzerland

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Research Interests: LLM interpretability, controllability, reasoning, human-centred AI, alignment, small language models

Education

École Polytechnique Fédérale de Lausanne (EPFL),
Ph.D. Computer Science

Lausanne, Switzerland
September 2024 – Present

- Supervised by Prof. Tanja Käser at ML4ED Lab
- Working on Interpretability and controllability of transformer models (LLMs/VLMs) with a focus on human-centric domains.

University College London (UCL),
M.Sc. in Data Science and Machine Learning

London, United Kingdom
September 2022

- Overall grade: Distinction
- Thesis Title: Optimisation of real time business transactions through intelligent agents system. Grade: Distinction

Selected Publications

REFINE: Real-world Exploration of Interactive Feedback and Student Behaviour
Fares Fawzi, Seyed Parsa Neshaei, Marta Knezevic, Tanya Nazaretsky, Tanja Käser
27th International Conference on Artificial Intelligence in Education **AIED Full Paper (Oral)**, 2026

SCRIBE: Structured Chain Reasoning for Interactive Behaviour Explanations using Tool Calling
Fares Fawzi, Vinitra Swamy, Dominik Glandorf, Tanya Nazaretsky, Tanja Käser
The 2025 Conference on Empirical Methods in Natural Language Processing **EMNLP Main**, 2025

Towards human-like educational question generation with small language models
Fares Fawzi, Sarang Balan, Mutlu Cukurova, Emine Yilmaz, Sahan Bulathwela
International Conference on Artificial Intelligence in Education **AIED**, 2024

Small generative language models for educational question generation
Fares Fawzi, Sadie Amini, Sahan Bulathwela
Proceedings of the **NeurIPS** Workshop on Generative Artificial Intelligence for Education (GAIEd), 2023

Research Experience

EPFL, ML4ED Lab, Ph.D. Researcher

Lausanne, Switzerland
September 2024 – Present

- Using **mechanistic interpretability** and feature circuits to uncover how LLMs internally represent and execute pedagogical moves, enabling transparent teacher-controllable AI tutors and shedding light on what language models learn about human tutoring.
- Extended **SCRIBE** to **REFINE** (AIED 2026), a multi-agent system built on **small open-source LLMs**, introducing **inference-time alignment** via judge-guided regeneration and structured tool-augmented reasoning. Deployed with **real undergraduate students**, we performed **behavioural interaction analyses** demonstrating that our iterative alignment improves feedback grounding and reasoning efficiency in authentic use.
- Developed **SCRIBE** (EMNLP 2025), a framework for structured multi-hop reasoning and tool-calling in small LLMs, enabling interactive and pedagogically grounded feedback.

UCL, AI Centre, Research Assistant

- Developed **EduQG**, a question generation system trained by pre-training T5-small on the *S2ORC* dataset and fine-tuning on *SQuAD*, achieving competitive quality with GPT-3.5 for educational QA.
- Designed and conducted a **human evaluation study** assessing pedagogical relevance and linguistic quality of generated questions.
- Published at **NeurIPS'23 Workshop on Generative AI for Education (GAIED)** and **AIED 2024**.

London, United Kingdom
August 2023 – July 2024

Industry Experience

InformedAI, Machine Learning Engineer

London, United Kingdom
Oct 2022 – Aug 2024

- Built a **multi-agent system for automating marketing operations**, leveraging **LLM function calling** to coordinate agents for customer profiling, content generation, and A/B testing automation.
- Developed an **e-commerce chatbot** using an information retrieval pipeline combining intent classification and semantic search, achieving 100% top-3 accuracy in product recommendations.
- **Fine-tuned open-source LLMs** (StableLM, GPT-2) for domain-specific text generation.
- Built a **federated learning framework** on AWS EC2 using **Flower** to train recommendation models without accessing private user data.

UCL & InformedAI, M.Sc. Dissertation Project (Distinction)

London, United Kingdom
Jun 2022 – Sep 2022

- Developed a **multi-agent recommender system** that autonomously optimised real-time business transactions.
- Designed three collaborating agents (Recommendation, Pricing, and Discount) communicating via AWS AmazonMQ.
- Trained a **TD3 reinforcement learning model** for pricing policy optimisation, achieving convergence and stability in simulation.
- Built gradient boosting and content-based models for eligibility prediction and car recommendation (98% accuracy, 83% mAP).

Awards and Achievements

SwissAI Initiative Fellowship, Computer and Communication Sciences Doctoral Program (EDIC), EPFL — 2025 – Present

EPFL EDIC Doctoral Fellowship, Computer and Communication Sciences Doctoral Program (EDIC), EPFL — 2024 – 2025

Double Major with Highest Honours, B.Sc. Mechanical Engineering & B.Sc. Physics — *Summa Cum Laude - Dean's List* — June 2021

1st Place Worldwide, Robotics and Design Competition, IDETC Conference – ASME — August 2019

Skills

Coding	Python, PyTorch, TensorFlow, CUDA, Git, Docker, Linux, Slurm, AWS
Deep Learning	Large Scale training Parallel & Distributed Computing (GPU Acceleration), Autoregressive Models
Generative AI	Large Language models (LLMs), Interpretability, Alignment, GPRO, RLHF, finetuning, human-centric AI, vision language models (VLMs), model distillation, RAG
Languages	English (Native), Arabic (Fluent), French (A2 written and spoken)

References

Tanja Käser, Professor at EPFL and PhD supervisor, tanja.kaeser@epfl.ch